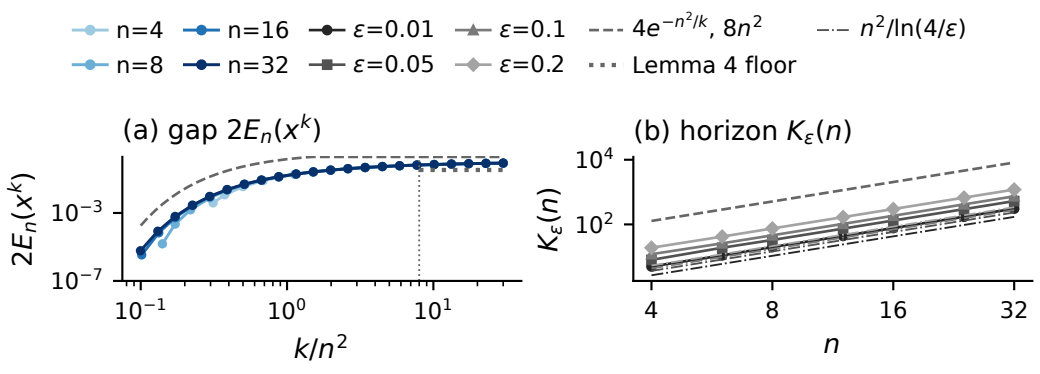
n=4
n=8
n=16
n=32
ε=0.01
ε=0.05
ε=0.1
ε=0.2
4e^{-n^2/k}, 8n^2
n^2/ln(4/ε)
Lemma 4 floor
(a) gap 2E_n(x^k)
2E_n(x^k)
10^{-3}
10^{-7}
10^{-1}
10^{0}
10^{1}
k/n^2
(b) horizon K_ε(n)
K_ε(n)
10^4
10^2
4
8
16
32
n